

CLIPTI MVP

AI-Powered NFC Clip Button for Streamers & Podcasters

Product Blueprint — v1.0

1. Executive Summary

****What It Is:**** A physical NFC "mark" button that lets streamers and podcasters instantly clip moments in real-time while streaming or recording. Connects to cameras (GoPro, iPhone, mirrorless) and sends clips to a desktop dashboard (Tauri) where AI auto-edits, captions, and queues content for posting.

****The Problem:****

- Streamers miss viral moments because they're focused on performing
- Manual clipping is time-consuming and requires post-production
- Existing solutions (Stream Deck) are generic, not purpose-built for clipping
- AI clipping tools (Opus Clip, Munch) work on recorded video only—not live

****The Solution:****

- One-tap NFC button to "mark" the moment
- Real-time buffer capture (last 30 seconds automatically saved)
- AI auto-generates clip with captions + hooks
- Desktop dashboard (Tauri) for review, editing, and scheduling

****Market Opportunity:****

- AI Hardware Market: \$833.4B by 2033 (31.5% CAGR)
- Edge AI Hardware: \$23.8B in 2024 → \$87.9B by 2032
- Creator Economy: \$250B+ market, 50M+ creators globally
- Stream Deck Alternatives: \$20-\$200+ segment, underserved for clipping

****Competition:****

- Stream Deck (\$150+) — Generic, not AI-native
- Opus Clip / Munch — Software only, no hardware
- DIY ESP32 projects — No polish, no integration

****Our Moat:****

- Hardware + Software + AI in one tight loop
- Real-time clip marking (not just recording)
- Built for creators, not enterprises

2. Market Analysis

2.1 Total Addressable Market

Segment	Market Size	Growth
Streamers (Twitch, YouTube, TikTok)	10M+ active	15% YoY
Podcasters	5M+ active	20% YoY
Video Podcasters	1M+ active	35% YoY
Total TAM	\$5B+	25% CAGR

2.2 Target Customer Profile

Attribute	Description
Primary	Streamers 1K-100K followers, gaming/entertainment
Secondary	Podcasters with video, 500-50K subscribers
Tertiary	Educators/course creators, corporate events
Pain Point	Missing viral moments, slow post-production
Willing to Pay	\$50-200 for time-saving hardware

2.3 Competitive Landscape

Competitor	Price	Strengths	Weaknesses
Elgato Stream Deck	\$150+	Ecosystem, reliability	No AI, generic buttons
Loupedeck Live S	\$199	Professional audio integration	Complex UI, expensive
Touch Portal (Software)	Free-\$50	Cheap, customizable	No hardware, unstable
DIY ESP32 Button	\$20-50	Cheap	No polish, DIY required
Opus Clip (Software)	\$25/mo	Great AI clipping	Post-production only

Gap: No one makes a purpose-built, AI-enabled clip button for live creators.

3. Product Concept

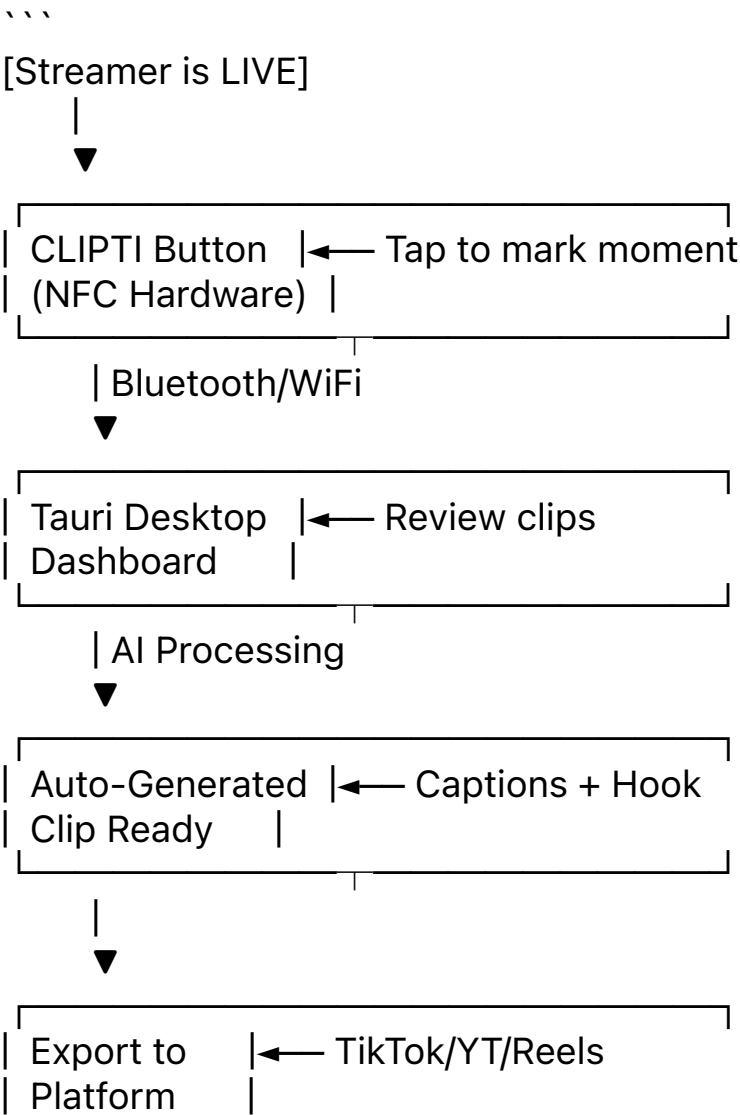
3.1 Product Name

****CLIPTI**** — "Clip" + "Ti" (touch/tap)

3.2 Core Features

Feature	Description	Priority
NFC Tap-to-Clip	One-tap marks the last 30 seconds	Must Have
Real-Time Buffer	Rolling 30s buffer always recording	Must Have
Camera Integration	GoPro, iPhone, USB camera support	Must Have
AI Auto-Caption	Whisper transcription + caption overlay	Must Have
Desktop Dashboard	Tauri app for review, edit, schedule	Must Have
Multi-Platform Export	TikTok, YouTube Shorts, Reels	Should Have
Stream Integration	Twitch, Kick, YouTube Live awareness	Could Have
Viral Prediction	AI scores clip potential	Could Have

3.3 User Flow



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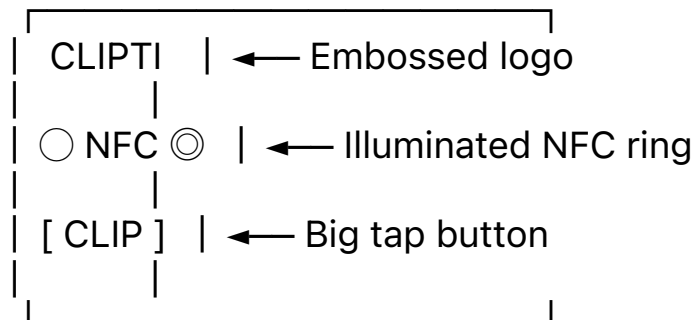
4. Hardware Design

4.1 Prototype: ESP32 + NFC Module

Component	Part	Cost (1K units)	Notes
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Main MCU	ESP32-S3	\$3.50	WiFi + BT, AI capable
NFC Module	PN532	\$2.00	Contactless payments-grade
Flash Storage	8MB SPI	\$0.80	Local clip buffer
Battery	500mAh LiPo	\$2.50	8h battery life
Enclosure	3D Printed	\$1.50	Custom shape
Button	Illuminated	\$1.00	RGB feedback
Total BOM		**\$11.30**	

4.2 Industrial Design

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USB-C Bluetooth

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Form Factor Options:

1. **Desktop Orb** — Sits on desk, tap to mark
2. **Wrist Fob** — Wearable, tap while gaming
3. **Pedal** — Foot tap for hands-free
4. **Mountable** — Attach to desk/mic stand

4.3 Connectivity

Protocol	Use Case	Range
Bluetooth 5.0	Desktop connection	10m
WiFi (AP Mode)	Direct camera connection	30m
USB-C	Data transfer + charging	Wired
BLE Long Range	Outdoor/扩展	range 100m+

5. Software Architecture

5.1 Desktop App: Tauri + React

Layer	Technology	Purpose
Frontend	React + TypeScript	UI dashboard
Framework	Tauri 2.0	Native desktop app
State	Zustand	Clip queue management
Styling	Tailwind CSS	Modern UI
Video	FFmpeg (Rust bindings)	Processing pipeline
AI	Local Whisper + DeepSeek	Transcription + hooks
Storage	SQLite	Clip metadata
Export	yt-dlp + platform APIs	Direct posting

5.2 Desktop Dashboard UI

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CLIPTI Dashboard										[Settings] [?]									
CLIPS					CLIP PREVIEW														
QUEUE																			
○ Clip 1																			
0:32					[Video Preview]														
○ Clip 2																			
0:45																			

○ Clip 3

0:28

Title: "Epic Gaming Moment #47"
Caption: "When your teammate ints but you..."
#gaming #viral #clipti
[Edit] [Add Captions] [Schedule] [Export]

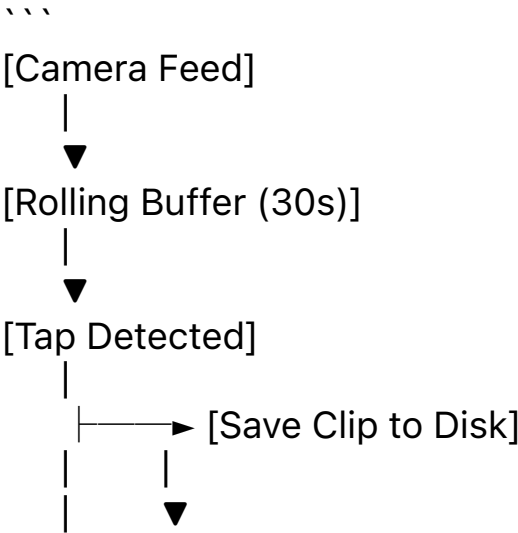
Connected: CLIPTI-001 | GoPro #1 | iPhone | Storage: 2.3GB

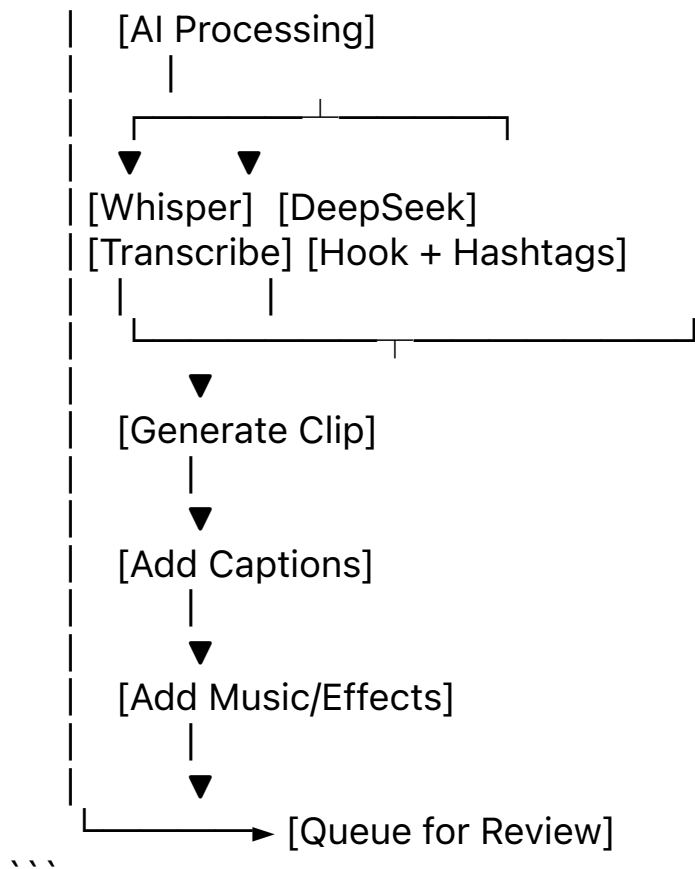
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5.3 Key Features

Feature	Implementation
Real-Time Buffer	FFmpeg captures last 30s rolling buffer
AI Transcription	Whisper.cpp (local, no cloud)
Caption Overlay	FFmpeg burn-in captions
Hook Generation	DeepSeek API call (optional local)
Platform Export	yt-dlp for upload, API integration
Auto-Posting	Schedule queue → cron → upload

5.4 Clip Processing Pipeline





6. Camera Integration

6.1 GoPro Integration

Method	Protocol	Status
GoPro API	WiFi HTTP	Supported (Hero 9+)
GoPro Media Mod	USB-C	Planned
RTMP Stream	WiFi	Supported

```
```python
Example: GoPro Clip Capture
import requests

Start GoPro recording
requests.get("http://10.5.5.9/gp/gpControl/execute?p1=gpStream&a1=proto_v2")

When CLIPTI tapped, save last 30 seconds from stream buffer
Process with FFmpeg
```

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### 6.2 iPhone Integration

Method	Protocol	Status
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**DOGCam SDK**	WiFi	Supported
**Capture App**	Bluetooth	Planned
**Continuity Camera**	USB/WiFi	Planned

### 6.3 USB Camera / Capture Card

Method	Protocol	Status
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**Direct USB**	UVC	Supported
**Capture Card**	HDMI → USB	Supported
**Blackmagic**	Thunderbolt	Planned

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## 7. Technical Specifications

### 7.1 Hardware Specs (MVP)

Specification	Value
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**MCU**	ESP32-S3-WROOM-1
**Clock Speed**	240 MHz
**RAM**	512KB
**Flash**	8MB
**Battery**	500mAh LiPo
**Battery Life**	8 hours active
**Connectivity**	BT 5.0 + WiFi 802.11n
**NFC Range**	0-5cm
**Dimensions**	60mm × 60mm × 20mm
**Weight**	45g
**Enclosure**	3D printed PLA/TPU
**Retail Price (Target)**	\$49-79

### 7.2 Software Requirements

Component	Version	Notes
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	**Tauri**   2.0+   Desktop framework		
	**React**   18+   Frontend UI		
	**TypeScript**   5.0+   Type safety		
	**FFmpeg**   6.0+   Video processing		
	**Whisper**   cpp / transformers   Local transcription		
	**Rust**   1.70+   Tauri backend		
	**SQLite**   3.x   Local database		
	**Node.js**   18+   Build tooling		

### 7.3 Desktop App Specs

	Requirement	Value
	-----	-----
	**OS Support**   Windows 10+, macOS 12+, Ubuntu 22.04+	
	**RAM**   4GB minimum, 8GB recommended	
	**Storage**   10GB+ for clips	
	**GPU**   Not required (AI runs CPU)	
	**Install Size**   ~100MB	

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## 8. Business Model

### 8.1 Revenue Streams

	Stream	Model	Price	Notes
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	**Hardware Sales**   One-time   \$79   MVP device			
	**Software Subscription**   Monthly   \$9.99/mo   AI features, cloud sync			
	**Enterprise**   Per seat   \$29/mo   Team features, API			
	**White Label**   OEM   Custom   Camera/microphone brands			

### 8.2 Pricing Tiers

	Tier	Price	Features
	-----	-----	-----
	**Free**   \$0   Basic clipping, local storage		
	**Pro**   \$9.99/mo   AI captions, hooks, cloud sync		
	**Studio**   \$29/mo   Multi-camera, team features		
	**Enterprise**   Custom   API, SSO, dedicated support		

### ### 8.3 Unit Economics

Metric	Value
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<b>BOM Cost</b>	\$11.30
<b>Assembly</b>	\$5.00
<b>Shipping</b>	\$8.00
<b>Total COGS</b>	\$24.30
<b>Retail Price</b>	\$79.00
<b>Gross Margin</b>	69%

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## ## 9. Roadmap

### ### Phase 1: MVP (Month 1-3)

- ☐ ESP32 prototype hardware
- ☐ Basic NFC tap detection
- ☐ BLE connection to desktop
- ☐ Tauri dashboard shell
- ☐ Rolling buffer capture (30s)
- ☐ Basic export (save to disk)

### ### Phase 2: Beta (Month 4-6)

- ☐ GoPro integration
- ☐ AI transcription (Whisper)
- ☐ Caption overlay
- ☐ 100 beta testers
- ☐ Feedback iteration

### ### Phase 3: Launch (Month 7-9)

- ☐ Production hardware
- ☐ Pre-order campaign (Kickstarter/Indiegogo)
- ☐ Ship to backers
- ☐ macOS + Windows app
- ☐ Basic platform export

### ### Phase 4: Scale (Month 10-12)

- ☐ iPhone integration
- ☐ Auto-post to TikTok/YT
- ☐ Hook generation (DeepSeek)
- ☐ Mobile companion app

- [ ] Enterprise features

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## ## 10. Go-to-Market Strategy

### ### 10.1 Launch Channels

Channel	Strategy	Expected
**Kickstarter**	Hardware campaign	\$100K-500K
**Product Hunt**	Launch day	500-2000 signups
**Reddit**	r/streamers, r/podcasting	Organic buzz
**TikTok**	Demo videos	Viral potential
**YouTube**	Reviews, tutorials	SEO long-tail

### ### 10.2 Target Influencers

Tier	Criteria	Count
**Nano**	1K-10K followers	50
**Micro**	10K-100K followers	20
**Macro**	100K-1M followers	5
**Mega**	1M+ followers	1-2

### ### 10.3 Early Adopter Program

Program	Price	Includes
**Early Bird**	\$49	First 500 units
**Founder's Edition**	\$69	Special packaging + lifetime Pro
**Developer Edition**	\$99	Hardware + API access

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## ## 11. Risk Analysis

Risk	Probability	Impact	Mitigation
**Hardware delays**	Medium	High	Use existing modules, avoid custom PCB
**BLE instability**	Low	Medium	USB fallback, robust pairing
**Camera API changes**	Medium	Medium	Support multiple methods

**Competition**	Medium	Low	First-mover in this category
**AI costs**	Low	Low	Local Whisper, cloud optional
**Supply chain**	Medium	High	Multiple component suppliers

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## 12. Technical References

### 12.1 Key Technologies

Technology	Use Case	Link
**Tauri**	Desktop app framework	<https://tauri.app>
**CrabCamera**	Tauri camera plugin	[crates.io/crates/crabcamera](https://crates.io/crates/crabcamera)
**Whisper.cpp**	Local transcription	[github.com/ggerganov/whisper.cpp](https://github.com/ggerganov/whisper.cpp)
**ESP-IDF**	ESP32 firmware	<https://docs.espressif.com>
**PN532**	NFC module	Adafruit PN532
**yt-dlp**	Video download/upload	[github.com/yt-dlp/yt-dlp](https://github.com/yt-dlp/yt-dlp)
**DeepSeek**	Hook generation	[platform.deepseek.com](https://platform.deepseek.com)

### 12.2 GitHub Repos to Reference

Repo	Purpose
[mvanhorn/last30days-skill](#)	Research intelligence
[tauri-apps/tauri](#)	Desktop framework
[ggerganov/whisper.cpp](#)	Local AI transcription
[OpusClip/opus-api](#)	Clip optimization

### 12.3 Similar Products

Product	Price	Notes
**Elgato Stream Deck**	\$149+	Generic buttons
**Loupedeck Live S**	\$199	Audio focus
**Yee虎 Button**	\$30	China-only
**DIY Stream Deck**	\$20-50	No AI, DIY

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## 13. Next Steps

### Immediate (Week 1-2)

- 1. Order ESP32 + PN532 dev kits
- 2. Set up Tauri project scaffolding
- 3. Implement BLE communication
- 4. Build rolling buffer prototype

### Short-term (Week 3-6)

- 1. Integrate GoPro API
- 2. Add Whisper transcription
- 3. Build dashboard UI
- 4. User testing with 5-10 creators

### Medium-term (Month 2-3)

- 1. PCB design for v1
- 2. 3D print enclosure
- 3. Pre-order campaign setup
- 4. Manufacturing quote

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## 14. Success Metrics

Metric   Target (6 months)
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**Pre-orders**   1,000 units
**MRR**   \$10K
**Active Users**   500
**NPS**   50+
**Retention**   60% (6 months)
**Clips Created**   100K+

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## 15. Conclusion

**CLIPTI** fills a clear gap in the creator hardware market: a purpose-built, AI-enabled clip button that's faster and smarter than a Stream Deck and more integrated than software-only solutions.

The timing is right:

- AI transcription is now local and free
- Creator economy is mature and growing

- Hardware costs have dropped (ESP32 is \$3.50)
- Tauri makes desktop apps fast and small

The business model is proven:

- Hardware + SaaS (Stream Deck, Loupedeck)
- Creator tools market is \$5B+
- Pre-order campaigns work for hardware

**\*\*The MVP is buildable in 3 months with \$10K budget.\*\***

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**\*Blueprint created: July 2026\***

**\*Author: ClipForge Labs\***